

Assignment(Document + Kaggle dataset(description/Question))

1) Load Superstore dataset into a table (superstore_raw).(step-1 from the document)

```
mysql> create table superstoreraw as select * from superstore;
Query OK, 9994 rows affected (0.16 sec)
Records: 9994 Duplicates: 0 Warnings: 0

mysql> select count(*) as totalrows from superstoreraw;
+-----+
| totalrows |
+-----+
|      9994 |
+-----+
1 row in set (0.01 sec)
```

2) Create tables (customers, orders, products) from the dataset.(step-1)

```
mysql> create table customers as select distinct customer_id, customer_name, segment, city, state, region from superstoreraw;
Query OK, 4688 rows affected (0.29 sec)
Records: 4688 Duplicates: 0 Warnings: 0
```

```
mysql> create table products as select distinct product_id, product_name, category, sub_category from superstoreraw;
Query OK, 1894 rows affected (0.06 sec)
Records: 1894 Duplicates: 0 Warnings: 0
```

```
mysql> create table orders as select distinct order_id, customer_id, order_date, ship_date, ship_mode, sales, quantity, discount, profit, product_id from superstoreraw;
Query OK, 9993 rows affected (0.13 sec)
Records: 9993 Duplicates: 0 Warnings: 0
```

```
mysql> select 'superstoreraw' as tbl, count(*) as row_count from superstoreraw union all select 'customers', count(*) from customers union all select 'products', count(*) from products union all select 'orders', count(*) from orders;
+-----+-----+
| tbl          | row_count |
+-----+-----+
| superstoreraw |      9994 |
| customers     |      4688 |
| products      |      1894 |
| orders        |      9993 |
+-----+-----+
4 rows in set (0.11 sec)
```

3) Insert data into these tables using SELECT DISTINCT.(step-1)

```
mysql> insert into customers select distinct customer_id, customer_name, segment, city, state, region from superstore;
Query OK, 4688 rows affected (0.32 sec)
Records: 4688 Duplicates: 0 Warnings: 0
```

```
mysql> insert into products select distinct product_id, product_name, category, sub_category from superstore;
Query OK, 1894 rows affected (0.05 sec)
Records: 1894 Duplicates: 0 Warnings: 0
```

```
mysql> insert into orders select distinct order_id, customer_id, order_date, ship_date, ship_mode, sales, quantity, discount, profit, product_id from superstore;
Query OK, 9993 rows affected (0.28 sec)
Records: 9993 Duplicates: 0 Warnings: 0
```

4) Apply subqueries to filter data (above average sales, highest order per customer).

```
mysql> with customer_sales as (select o.customer_id, c.customer_name, round(sum(o.sales),2) as total_sales, round(sum(o.profit),2) as total_profit, count(distinct o.order_id) as total_orders from orders o join customers c on o.customer_id = c.customer_id group by o.customer_id, c.customer_name) select * from customer_sales order by total_sales desc;
```

customer_id	customer_name	total_sales	total_profit	total_orders
KL-1664512	Ken Lonsdale	567009.16	32274.20	12
SE-2011011	Sanjit Engle	537215.27	116629.78	11
AB-1010510	Adrian Barton	521048.56	196013.00	10
SM-203205	Sean Miller	500861.00	-39614.79	5
CL-1256512	Clay Ludtke	478744.02	85086.46	12
SC-200959	Sanjit Chand	452554.69	184237.18	9
SV-20365	Seth Vernon	412954.20	43179.27	1

```
mysql> select order_id, customer_id, product_id, round(sales,2) as sales, round((select avg(sales) from orders),2) as avg_sales from orders where sales > (select avg(sales) from orders) order by sales desc;
```

order_id	customer_id	product_id	sales	avg_sales
CA-2014-145317	SM-20320	TEC-MA-10002412	22638.48	229.85
CA-2016-118689	TC-20980	TEC-CO-10004722	17499.95	229.85
CA-2017-140151	RB-19360	TEC-CO-10004722	13999.96	229.85
CA-2017-127180	TA-21385	TEC-CO-10004722	11199.97	229.85
CA-2017-166709	HL-15040	TEC-CO-10004722	10499.97	229.85
CA-2016-117121	AB-10105	OFF-BI-10000545	9892.74	229.85
CA-2014-116904	SC-20095	OFF-BI-10001120	9449.95	229.85
US-2016-107440	BS-11365	TEC-MA-10001047	9099.93	229.85
CA-2016-158841	SE-20110	TEC-MA-10001127	8749.95	229.85
CA-2016-143714	CC-12370	TEC-CO-10004722	8399.98	229.85
CA-2014-143917	KL-16645	OFF-SU-10000151	8187.65	229.85
CA-2014-139892	BM-11140	TEC-MA-10000822	8159.95	229.85
US-2017-168116	GT-14635	TEC-MA-10004125	7999.98	229.85
CA-2014-145541	TB-21400	TEC-MA-10001127	6999.96	229.85
CA-2015-145352	CM-12385	OFF-BI-10003527	6354.95	229.85
CA-2017-138289	AR-10540	OFF-BI-10004995	5443.96	229.85
US-2016-140158	DR-12940	TEC-CO-10001449	5399.91	229.85
CA-2017-143112	TS-21370	TEC-MA-10001047	5199.96	229.85

```
mysql> select o.customer_id, c.customer_name, o.order_id, round(o.sales,2) as top_order_sales from orders o join customers c on o.customer_id = c.customer_id where (o.customer_id, o.sales) in (select customer_id, max(sales) from orders group by customer_id) order by top_order_sales desc limit 20;
```

customer_id	customer_name	order_id	top_order_sales
SM-20320	Sean Miller	CA-2014-145317	22638.48
SM-20320	Sean Miller	CA-2014-145317	22638.48
SM-20320	Sean Miller	CA-2014-145317	22638.48
SM-20320	Sean Miller	CA-2014-145317	22638.48
SM-20320	Sean Miller	CA-2014-145317	22638.48
TC-20980	Tamara Chand	CA-2016-118689	17499.95
TC-20980	Tamara Chand	CA-2016-118689	17499.95
TC-20980	Tamara Chand	CA-2016-118689	17499.95
TC-20980	Tamara Chand	CA-2016-118689	17499.95
TC-20980	Tamara Chand	CA-2016-118689	17499.95
RB-19360	Raymond Buch	CA-2017-140151	13999.96
RB-19360	Raymond Buch	CA-2017-140151	13999.96
RB-19360	Raymond Buch	CA-2017-140151	13999.96
RB-19360	Raymond Buch	CA-2017-140151	13999.96
RB-19360	Raymond Buch	CA-2017-140151	13999.96
RB-19360	Raymond Buch	CA-2017-140151	13999.96
TA-21385	Tom Ashbrook	CA-2017-127180	11199.97
TA-21385	Tom Ashbrook	CA-2017-127180	11199.97
TA-21385	Tom Ashbrook	CA-2017-127180	11199.97
HL-15040	Hunter Lopez	CA-2017-166709	10499.97

```
mysql> select customer_id, round(sum(sales),2) as total_sales from orders group by customer_id having sum(sales) > (select avg(total) from (select sum(sales) as total from orders group by customer_id) as sub) order by total_sales desc;
```

customer_id	total_sales
SM-20320	25043.05
TC-20980	19052.22
RB-19360	15117.34
TA-21385	14595.62
AB-10105	14473.57
KL-16645	14175.23
SC-20095	14142.33
HL-15040	12873.30
SE-20110	12209.44
CC-12370	12129.07
TS-21370	11891.75
GT-14710	11820.12
BM-11140	11789.63
SV-20365	11470.95
CJ-12010	11164.97
CL-12565	10880.55
ME-17320	10663.73
KF-16285	10604.27

```
mysql> select distinct product_id, product_name from products where product_id not in (select distinct product_id from orders where discount > 0);
```

product_id	product_name
FUR-FU-10001487	Eldon Expressions Wood and Plastic Desk Accessories, Cherry Wood
OFF-ST-10004186	Stur-D-Stor Shelving, Vertical 5-Shelf: 72"H x 36"W x 18 1/2"D
OFF-LA-10002762	Avery 485
FUR-FU-10000397	Luxo Economy Swing Arm Lamp
FUR-FU-10001934	Magnifier Swing Arm Lamp
FUR-FU-10002157	Artistic Insta-Plaque
TEC-PH-10003875	KLD Oscar II Style Snap-on Ultra Thin Side Flip Synthetic Leather Cover Case for HTC One HTC M7
OFF-SU-10001218	Fiskars Softgrip Scissors
OFF-PA-10002751	Xerox 1920
OFF-EN-10001990	Staple envelope
OFF-ST-10001414	Decoflex Hanging Personal Folder File
OFF-FA-10000134	Advantus Push Pins, Aluminum Head

5) Use CTEs to compute aggregations (total sales per customer).

```
mysql> with customer_sales as (select o.customer_id, c.customer_name, c.segment, c.region, round(sum(o.sales),2) as total_sales, round(sum(o.profit),2) as total_profit, count(distinct o.order_id) as total_orders from orders o join customers c on o.customer_id = c.customer_id group by o.customer_id, c.customer_name, c.segment, c.region) select * from customer_sales order by total_sales desc;
```

customer_id	customer_name	segment	region	total_sales	total_profit	total_orders
SE-20110	Sanjit Engle	Consumer	West	85466.07	18554.74	11
AB-10105	Adrian Barton	Consumer	Central	72367.86	27224.03	10
KL-16645	Ken Lonsdale	Consumer	West	70876.15	4034.28	12
SV-20365	Seth Vernon	Consumer	East	68825.70	7196.55	10
SC-20095	Sanjit Chand	Consumer	West	56569.34	23029.65	9
SM-20320	Sean Miller	Home Office	South	50086.10	-3961.48	5
CJ-12010	Caroline Jumper	Consumer	East	44659.90	3434.97	8
CL-12565	Clay Ludtke	Consumer	East	43522.18	7735.13	12
CL-12565	Clay Ludtke	Consumer	West	43522.18	7735.13	12

```
mysql> with customer_sales as (select o.customer_id, c.customer_name, round(sum(o.sales),2) as total_sales from orders o join customers c on o.customer_id = c.customer_id group by o.customer_id, c.customer_name) select * from customer_sales order by total_sales desc limit 10;
```

customer_id	customer_name	total_sales
KL-16645	Ken Lonsdale	141752.29
SE-20110	Sanjit Engle	134303.82
AB-10105	Adrian Barton	130262.14
SM-20320	Sean Miller	125215.25
CL-12565	Clay Ludtke	119686.01
SC-20095	Sanjit Chand	113138.67
SV-20365	Seth Vernon	103238.55
ZC-21910	Zuschuss Carroll	96308.48
ME-17320	Maria Etezadi	95973.55
LA-16780	Laura Armstrong	95405.44

10 rows in set (0.13 sec)

```
mysql> with customer_sales as (select o.customer_id, c.customer_name, round(
sum(o.sales),2) as total_sales from orders o join customers c on o.customer_
id = c.customer_id group by o.customer_id, c.customer_name) select * from cu
stomer_sales order by total_sales asc limit 10;
```

customer_id	customer_name	total_sales
LD-16855	Lela Donovan	5.30
TS-21085	Thais Sissman	9.67
CJ-11875	Carl Jackson	16.52
MG-18205	Mitch Gastineau	16.74
RS-19870	Roy Skaria	44.66
RE-19405	Ricardo Emerson	48.36
AR-10570	Anemone Ratner	88.15
RM-19750	Roland Murray	98.35
AS-10135	Adrian Shami	117.64
SG-20890	Susan Gilcrest	143.84

10 rows in set (0.12 sec)

```
mysql> with category_stats as (select p.category, round(sum(o.sales),2) as t
otal_sales, round(sum(o.profit),2) as total_profit, round(avg(o.sales),2) as
avg_sales, count(distinct o.order_id) as total_orders, round((sum(o.profit)
/sum(o.sales))*100,2) as profit_margin_pct from orders o join products p on
o.product_id = p.product_id group by p.category) select * from category_stat
s order by total_sales desc;
```

category	total_sales	total_profit	avg_sales	total_orders	profit_margin_pct
Technology	893633.28	153415.70	463.26	1544	17.17
Furniture	764003.28	20110.95	344.92	1764	2.63
Office Supplies	736748.59	126113.35	119.10	3742	17.12

3 rows in set (0.04 sec)

6) Apply window functions (ROW_NUMBER, RANK) for ranking and analysis.

```
mysql> select customer_id, order_id, round(sales,2) as sales, row_number() o
ver (partition by customer_id order by sales desc) as row_num from orders or
der by customer_id, row_num limit 50;
```

customer_id	order_id	sales	row_num
AA-10315	CA-2016-103982	3930.07	1
AA-10315	CA-2014-128055	673.57	2
AA-10315	CA-2016-103982	431.98	3
AA-10315	CA-2017-147039	362.94	4
AA-10315	CA-2014-128055	52.98	5
AA-10315	CA-2016-103982	41.72	6
AA-10315	CA-2015-121391	26.96	7
AA-10315	CA-2014-138100	14.94	8
AA-10315	CA-2014-138100	14.56	9
AA-10315	CA-2017-147039	11.54	10
AA-10315	CA-2016-103982	2.30	11
AA-10375	CA-2016-131065	499.98	1
AA-10375	CA-2015-140921	149.97	2
AA-10375	CA-2017-100230	116.98	3
AA-10375	CA-2015-114503	84.96	4
AA-10375	US-2017-169488	39.96	5
AA-10375	CA-2014-130729	34.27	6
AA-10375	CA-2015-140921	28.40	7

```
mysql> select customer_id, round(sum(sales),2) as total_sales, rank() over (
order by sum(sales) desc) as sales_rank from orders group by customer_id order
by sales_rank limit 20;
```

customer_id	total_sales	sales_rank
SM-20320	25043.05	1
TC-20980	19052.22	2
RB-19360	15117.34	3
TA-21385	14595.62	4
AB-10105	14473.57	5
KL-16645	14175.23	6
SC-20095	14142.33	7
HL-15040	12873.30	8
SE-20110	12209.44	9
CC-12370	12129.07	10
TS-21370	11891.75	11
GT-14710	11820.12	12
BM-11140	11789.63	13
SV-20365	11470.95	14
CJ-12010	11164.97	15
CL-12565	10880.55	16

```
mysql> select product_id, round(sum(sales),2) as total_sales, dense_rank() o
ver (order by sum(sales) desc) as dense_rank_pos from orders group by produc
t_id order by dense_rank_pos limit 20;
```

product_id	total_sales	dense_rank_pos
TEC-CO-10004722	61599.82	1
OFF-BI-10003527	27453.38	2
TEC-MA-10002412	22638.48	3
FUR-CH-10002024	21870.58	4
OFF-BI-10001359	19823.48	5
OFF-BI-10000545	19024.50	6
TEC-CO-10001449	18839.69	7
TEC-MA-10001127	18374.90	8
OFF-BI-10004995	17965.07	9
OFF-SU-10000151	17030.31	10
TEC-MA-10000822	16829.90	11
OFF-SU-10002881	16656.20	12
OFF-BI-10001120	15875.92	13
FUR-BO-10004834	15610.97	14
TEC-MA-10001047	14299.89	15
TEC-PH-10001459	13943.67	16
TEC-AC-10002049	13756.54	17
TEC-PH-10002885	12996.60	18

```
mysql> select order_date, round(sales,2) as daily_sales, round(sum(sales) over (order by order_date rows between unbounded preceding and current row),2) as running_total from orders order by order_date limit 50;
```

order_date	daily_sales	running_total
2014-01-03	16.45	16.45
2014-01-04	11.78	28.23
2014-01-04	272.74	300.97
2014-01-04	3.54	304.51
2014-01-05	19.54	324.04
2014-01-06	19.44	343.48
2014-01-06	12.78	356.26
2014-01-06	2573.82	2930.08
2014-01-06	609.98	3540.06
2014-01-06	5.48	3545.54
2014-01-06	391.98	3937.52
2014-01-06	755.96	4693.48
2014-01-06	31.12	4724.60

```
mysql> select customer_id, order_id, order_date, round(sales,2) as sales, round(lag(sales) over (partition by customer_id order by order_date),2) as prev_order_sales, round(sales - lag(sales) over (partition by customer_id order by order_date),2) as sales_change from orders order by customer_id, order_date limit 50;
```

customer_id	order_id	order_date	sales	prev_order_sales	sales_change
AA-10315	CA-2014-128055	2014-03-31	52.98	NULL	
AA-10315	CA-2014-128055	2014-03-31	673.57	52.98	620.59
AA-10315	CA-2014-138100	2014-09-15	14.56	673.57	-659.01
AA-10315	CA-2014-138100	2014-09-15	14.94	14.56	0.38
AA-10315	CA-2015-121391	2015-10-04	26.96	14.94	12.02
AA-10315	CA-2016-103982	2016-03-03	2.30	26.96	-24.66
AA-10315	CA-2016-103982	2016-03-03	431.98	2.30	429.67
AA-10315	CA-2016-103982	2016-03-03	41.72	431.98	-390.26

```
mysql> with customer_sales as (select customer_id, round(sum(sales),2) as total_sales from orders group by customer_id) select customer_id, total_sales, ntile(4) over (order by total_sales desc) as quartile from customer_sales order by quartile, total_sales desc;
```

customer_id	total_sales	quartile
SM-20320	25043.05	1
TC-20980	19052.22	1
RB-19360	15117.34	1
TA-21385	14595.62	1
AB-10105	14473.57	1
KL-16645	14175.23	1
SC-20095	14142.33	1
HL-15040	12873.30	1
SE-20110	12209.44	1
CC-12370	12129.07	1
TS-21370	11891.75	1
GT-14710	11820.12	1
BM-11140	11789.63	1
SV-20365	11470.95	1
CJ-12010	11164.97	1
CL-12565	10880.55	1
ME-17320	10663.73	1
KF-16285	10604.27	1
BS-11365	10501.65	1
EH-13765	10310.88	1
JL-15835	9799.92	1
GT-14635	9351.21	1
HW-14935	9300.25	1
TB-21400	9133.99	1
PF-19120	9062.86	1
CM-12385	8954.02	1

7) Combine JOIN + CTE + Window Functions for final result (customer, total sales, rank).

```
mysql> with customer_sales as (select o.customer_id, c.customer_name, c.segment, c.region, round(sum(o.sales),2) as total_sales, round(sum(o.profit),2) as total_profit, count(distinct o.order_id) as total_orders from orders o join customers c on o.customer_id = c.customer_id group by o.customer_id, c.customer_name, c.segment, c.region), ranked_customers as (select customer_id, customer_name, segment, region, total_sales, total_profit, total_orders, rank() over (order by total_sales desc) as sales_rank, dense_rank() over (order by total_sales desc) as d_rank, round(total_sales / sum(total_sales) over () * 100,2) as pct_of_total from customer_sales) select * from ranked_customers order by sales_rank limit 20;
```

customer_id	customer_name	segment	region	total_sales	total_profit	total_orders	sales_rank	d_rank	pct_of_total
SE-20110	Sanjit Engle	Consumer	West	85466.07	18554.74	11	1	1	0.55
AB-10105	Adrian Barton	Consumer	Central	72367.86	27224.03	10	2	2	0.47
KL-16645	Ken Lonsdale	Consumer	West	70876.15	4034.28	12	3	3	0.46
SV-20365	Seth Vernon	Consumer	East	68825.70	7196.55	10	4	4	0.44
SC-20095	Sanjit Chand	Consumer	West	56569.34	23029.65	9	5	5	0.36
SM-20320	Sean Miller	Home Office	South	50086.10	-3961.48	5	6	6	0.32
CJ-12010	Caroline Jumper	Consumer	East	44659.90	3434.97	8	7	7	0.29
CL-12565	Clay Ludtke	Consumer	East	43522.18	7735.13	12	8	8	0.28
CL-12565	Clay Ludtke	Consumer	West	43522.18	7735.13	12	8	8	0.28
LA-16780	Laura Armstrong	Corporate	Central	43366.11	10295.60	11	10	9	0.28
ME-17320	Maria Fitzgerald	Home Office	Central	42654.01					

8) Solve business queries (top customers, low customers, single-order customers, above-average sales).

```
mysql> with customer_totals as (select o.customer_id, c.customer_name, c.segment, round(sum(o.sales),2) as total_sales, round(sum(o.profit),2) as total_profit from orders o join customers c on o.customer_id = c.customer_id group by o.customer_id, c.customer_name, c.segment) select customer_name, segment, total_sales, total_profit, rank() over (order by total_sales desc) as rank_pos from customer_totals order by rank_pos limit 10;
```

customer_name	segment	total_sales	total_profit	rank_pos
Ken Lonsdale	Consumer	141752.29	8068.55	1
Sanjit Engle	Consumer	134303.82	29157.45	2
Adrian Barton	Consumer	130262.14	49003.25	3
Sean Miller	Home Office	125215.25	-9903.70	4
Clay Ludtke	Consumer	119686.01	21271.61	5
Sanjit Chand	Consumer	113138.67	46059.30	6
Seth Vernon	Consumer	103238.55	10794.82	7
Zuschuss Carroll	Consumer	96308.48	-12385.79	8
Maria Etezadi	Home Office	95973.55	16735.23	9
Laura Armstrong	Corporate	95405.44	22650.32	10

10 rows in set (0.22 sec)

```
mysql> with customer_totals as (select o.customer_id, c.customer_name, round(sum(o.sales),2) as total_sales from orders o join customers c on o.customer_id = c.customer_id group by o.customer_id, c.customer_name) select customer_name, total_sales, rank() over (order by total_sales asc) as rank_pos from customer_totals order by rank_pos limit 10;
```

customer_name	total_sales	rank_pos
Lela Donovan	5.30	1
Thais Sissman	9.67	2
Carl Jackson	16.52	3
Mitch Gastineau	16.74	4
Roy Skaria	44.66	5
Ricardo Emerson	48.36	6
Anemone Ratner	88.15	7
Roland Murray	98.35	8
Adrian Shami	117.64	9
Susan Gilcrest	143.84	10

10 rows in set (0.12 sec)

```
mysql> select o.customer_id, c.customer_name, count(distinct o.order_id) as
order_count from orders o join customers c on o.customer_id = c.customer_id
group by o.customer_id, c.customer_name having count(distinct o.order_id) =
1 order by customer_name;
```

customer_id	customer_name	order_count
AR-10570	Anemone Ratner	1
AO-10810	Anthony O'Donnell	1
CJ-11875	Carl Jackson	1
JC-15385	Jenna Caffey	1
JR-15700	Jocasta Rupert	1
LD-16855	Lela Donovan	1
MG-18205	Mitch Gastineau	1
PH-18790	Patricia Hirasaki	1
RE-19405	Ricardo Emerson	1
RM-19750	Roland Murray	1
SM-20905	Susan MacKendrick	1
TC-21145	Theresa Coyne	1

12 rows in set (0.32 sec)

```
mysql> with avg_sales as (select avg(sales) as avg_val from orders) select o
.order_id, c.customer_name, c.region, p.category, round(o.sales,2) as order_
sales, round(a.avg_val,2) as overall_avg, round(o.sales - a.avg_val,2) as ab
ove_avg_by from orders o join customers c on o.customer_id = c.customer_id j
oin products p on o.product_id = p.product_id cross join avg_sales a where o
.sales > a.avg_val order by order_sales desc limit 20;
```

order_id	customer_name	region	category	order_sales	over all_avg	above_avg_by
CA-2014-145317	Sean Miller	West	Technology	22638.48	229.85	22408.63
CA-2014-145317	Sean Miller	East	Technology	22638.48	229.85	22408.63
CA-2014-145317	Sean Miller	South	Technology	22638.48	229.85	22408.63
CA-2014-145317	Sean Miller	South	Technology	22638.48	229.85	22408.63
CA-2014-145317	Sean Miller	Central	Technology	22638.48	229.85	22408.63
CA-2016-118689	Tamara Chand	South	Technology	17499.95		

```
mysql> with monthly as (select date_format(order_date,'%Y-%m') as month, round(sum(sales),2) as monthly_sales, round(sum(profit),2) as monthly_profit from orders group by date_format(order_date,'%Y-%m')) select month, monthly_sales, monthly_profit, round(sum(monthly_sales) over (order by month rows between unbounded preceding and current row),2) as cumulative_sales from monthly order by month;
```

month	monthly_sales	monthly_profit	cumulative_sales
2014-01	14236.90	2450.19	14236.90
2014-02	4519.89	862.31	18756.79
2014-03	55691.01	498.73	74447.80
2014-04	28013.97	3500.89	102461.77
2014-05	23648.29	2738.71	126110.06
2014-06	34595.13	4976.52	160705.19
2014-07	33946.39	-841.48	194651.58
2014-08	27909.47	5318.11	222561.05
2014-09	81777.35	8328.10	304338.40
2014-10	31453.39	3448.26	335791.79
2014-11	78628.72	9292.13	414420.51
2014-12	69545.62	8983.57	483966.13
2015-01	18174.08	-3281.01	502140.21
2015-02	11951.41	2813.85	514091.62
2015-03	38726.25	9732.10	552817.87
2015-04	34195.21	4187.50	587013.08
2015-05	30131.69	4667.87	617144.77
2015-06	24797.29	3335.56	641942.06
2015-07	28765.33	3288.65	670707.39

Step-(2)

1. orders where sales > average (subquery)

```
mysql> select * from orders where sales > (select avg(sales) from orders);
```

order_id	customer_id	order_date	ship_date	ship_mode	sales	quantity	discount	profit	product_id
CA-2016-152156	CG-12520	2016-11-08	2016-11-11	Second Class	261.9600	2	0.0000	41.9136	FUR-BO-10001798
CA-2016-152156	CG-12520	2016-11-08	2016-11-11	Second Class	731.9400	3	0.0000	219.5820	FUR-CH-10000454
US-2015-108966	SO-20335	2015-10-11	2015-10-18	Standard Class	957.5775	5	0.4500	-383.0310	FUR-TA-10000577
CA-2014-115812	BH-11710	2014-06-09	2014-06-14	Standard Class	907.1520	6	0.2000	90.7152	TEC-PH-10002275
CA-2014-115812	BH-11710	2014-06-09	2014-06-14	Standard Class	1706.1840	9	0.2000	85.3092	FUR-TA-10001539
CA-2014-115812	BH-11710	2014-06-09	2014-06-14	Standard Class	911.4240	4	0.2000	68.3568	TEC-PH-10002033
CA-2016-161389	IM-15070	2016-12-05	2016-12-10	Standard Class	407.9760	3	0.2000	132.5922	OFF-BI-10003656

2. Find the highest sales order for each customer. (Subquery)

```
mysql> select * from orders o where sales = (select max(sales) from orders w
here customer_id = o.customer_id);
```

order_id	customer_id	order_date	ship_date	ship_mode	sales	quantity	discount	profit	product_id
CA-2016-152156	CG-12520	2016-11-08	2016-11-11	Second Class	731.9400	3	0.0000	219.5820	FUR-CH-10000454
US-2015-108966	SO-20335	2015-10-11	2015-10-18	Standard Class	957.5775	5	0.4500	-383.0310	FUR-TA-10000577
CA-2014-115812	BH-11710	2014-06-09	2014-06-14	Standard Class	1706.1840	9	0.2000	85.3092	FUR-TA-10001539
CA-2015-106320	EB-13870	2015-09-25	2015-09-30	Standard Class	1044.6300	3	0.0000	240.2649	FUR-TA-10000577
US-2015-150630	TB-21520	2015-09-17	2015-09-21	Standard Class	3083.4300	7	0.5000	-1665.0522	FUR-BO-10004834
CA-2016-117590	GH-14485	2016-12-08	2016-12-10	First Class	1097.5440	7	0.2000	123.4737	TEC-PH-10004977
CA-2015-117415	SN-20710	2015-12-27	2015-12-31	Standard Class	532.3992	3	0.3200	-46.9764	FUR-BO-10002545
CA-2016-105816	JM-15265	2016-12-11	2016-12-17	Standard Class	1029.9500	5	0.0000	298.6855	TEC-PH-10002447
CA-2016-111682	TB-21055	2016-06-17	2016-06-18	First Class	319.4100	5	0.1000	7.0980	FUR-CH-10003968
CA-2014-106376	BS-11590	2014-12-05	2014-12-10	Standard Class					

3. Calculate total sales for each customer. (CTE)

```
mysql> with customer_sales as (select customer_id, sum(sales) as total_sales
from orders group by customer_id) select * from customer_sales;
```

customer_id	total_sales
CG-12520	2297.5600
DV-13045	2238.9660
SO-20335	5205.1510
BH-11710	12510.7020
AA-10480	3581.0240
IM-15070	9860.9480
HP-14815	3980.6280
PK-19075	17293.8680
AG-10270	5165.7960
ZD-21925	2987.8880
KB-16585	5489.4840
SF-20065	1527.0920
EB-13870	5534.4380
EH-13945	1368.3420
TB-21520	9474.9720
MA-17560	8598.3220
GH-14485	2722.4880
SN-20710	6255.9184
CG-16030	8885.8820

4. Find customers whose total sales are above average. (CTE + Subquery)

```
mysql> with customer_sales as (select customer_id, sum(sales) as total_sales
  from orders group by customer_id) select * from customer_sales where total_
sales > (select avg(total_sales) from customer_sales);
```

customer_id	total_sales
BH-11710	12510.7020
IM-15070	9860.9480
PK-19075	17293.8680
TB-21520	9474.9720
MA-17560	8598.3220
SN-20710	6255.9184
LC-16930	8985.8920
RA-19885	7664.6280
ES-14080	9315.8480
KM-16720	9818.9440
KD-16270	16564.7160
TE-15745	12521.6200

5. Rank all customers based on total sales. (Window Function)

```
mysql> with customer_sales as (select customer_id, sum(sales) as total_sales
  from orders group by customer_id) select customer_id, total_sales, rank() o
ver (order by total_sales desc) as rnk from customer_sales;
```

customer_id	total_sales	rnk
SM-20320	50086.1000	1
TC-20980	38104.4360	2
RB-19360	30234.6780	3
TA-21385	29191.2400	4
AB-10105	28947.1420	5
KL-16645	28350.4580	6
SC-20095	28284.6680	7
HL-15040	25746.5960	8
SE-20110	24418.8760	9
CC-12370	24258.1440	10
TS-21370	23783.5020	11
GT-14710	23640.2400	12
BM-11140	23579.2600	13

6. Assign row numbers to each order within a customer. (*Window Function + PARTITION BY*)

```
mysql> select order_id, customer_id, sales, row_number() over (partition by
customer_id order by sales desc) as row_num from orders;
```

order_id	customer_id	sales	row_num
CA-2016-103982	AA-10315	3930.0720	1
CA-2016-103982	AA-10315	3930.0720	2
CA-2014-128055	AA-10315	673.5680	3
CA-2014-128055	AA-10315	673.5680	4
CA-2016-103982	AA-10315	431.9760	5
CA-2016-103982	AA-10315	431.9760	6
CA-2017-147039	AA-10315	362.9400	7
CA-2017-147039	AA-10315	362.9400	8
CA-2014-128055	AA-10315	52.9800	9
CA-2014-128055	AA-10315	52.9800	10
CA-2016-103982	AA-10315	41.7200	11
CA-2016-103982	AA-10315	41.7200	12
CA-2015-121391	AA-10315	26.9600	13
CA-2015-121391	AA-10315	26.9600	14
CA-2014-138100	AA-10315	14.9400	15
CA-2014-138100	AA-10315	14.9400	16
CA-2014-138100	AA-10315	14.5600	17
CA-2014-138100	AA-10315	14.5600	18
CA-2017-147039	AA-10315	11.5400	19
CA-2017-147039	AA-10315	11.5400	20

7. Display top 3 customers based on total sales. (*Window Function*)

```
mysql> with customer_sales as (select customer_id, sum(sales) as total_sales
from orders group by customer_id) select * from (select customer_id, total_
sales, rank() over (order by total_sales desc) as rnk from customer_sales) t
where rnk <= 3;
```

customer_id	total_sales	rnk
SM-20320	50086.1000	1
TC-20980	38104.4360	2
RB-19360	30234.6780	3

3 rows in set (0.03 sec)

Step 3: Final Combined Query

Write one final query that shows:

- Customer Name
- Total Sales
- Rank

(Use JOIN + CTE + Window Function together)

```
mysql> with customer_sales as (select customer_id, sum(sales) as total_sales
from orders group by customer_id) select c.customer_name, cs.total_sales, r
ank() over (order by cs.total_sales desc) as rnk from customer_sales cs join
customers c on cs.customer_id = c.customer_id;
```

customer_name	total_sales	rnk
Sean Miller	50086.1000	1
Sean Miller	50086.1000	1
Sean Miller	50086.1000	1
Sean Miller	50086.1000	1
Sean Miller	50086.1000	1
Sean Miller	50086.1000	1
Sean Miller	50086.1000	1
Sean Miller	50086.1000	1
Sean Miller	50086.1000	1
Sean Miller	50086.1000	1
Tamara Chand	38104.4360	11
Tamara Chand	38104.4360	11
Tamara Chand	38104.4360	11
Tamara Chand	38104.4360	11
Tamara Chand	38104.4360	11
Tamara Chand	38104.4360	11
Tamara Chand	38104.4360	11
Tamara Chand	38104.4360	11
Tamara Chand	38104.4360	11
Tamara Chand	38104.4360	11
Raymond Buch	30234.6780	21

Mini Project: Customer Sales Insights

Answer the following using SQL:

- Who are the top 5 customers?

```
mysql> with cs as (select c.customer_id,c.customer_name,sum(o.sales) total_s
ales from customers c join orders o on c.customer_id=o.customer_id group by
c.customer_id,c.customer_name) select * from (select *,rank() over(order by
total_sales desc) rnk from cs) t where rnk<=5;
```

customer_id	customer_name	total_sales	rnk
KL-16645	Ken Lonsdale	567009.1600	1
SE-20110	Sanjit Engle	537215.2720	2
AB-10105	Adrian Barton	521048.5560	3
SM-20320	Sean Miller	500861.0000	4
CL-12565	Clay Ludtke	478744.0240	5

5 rows in set (2.42 sec)

- Who are the bottom 5 customers?

```
mysql> with cs as (select c.customer_id,c.customer_name,sum(o.sales) total_s
ales from customers c join orders o on c.customer_id=o.customer_id group by
c.customer_id,c.customer_name) select * from (select *,rank() over(order by
total_sales asc) rnk from cs) t where rnk<=5;
```

customer_id	customer_name	total_sales	rnk
LD-16855	Lela Donovan	21.2160	1
TS-21085	Thais Sissman	38.6640	2
CJ-11875	Carl Jackson	66.0800	3
MG-18205	Mitch Gastineau	66.9560	4
RS-19870	Roy Skaria	178.6240	5

5 rows in set (2.34 sec)

```
mysql>
```

3. Which customers made only one order?

```
mysql> select c.customer_id,c.customer_name from customers c join orders o o
n c.customer_id=o.customer_id group by c.customer_id,c.customer_name having
count(o.order_id)=1;
Empty set (1.99 sec)
```

4. Which customers have above-average sales?

```
mysql> with cs as (select customer_id,sum(sales) total_sales from orders gro
up by customer_id),avg_s as (select avg(total_sales) avg_val from cs) select
c.customer_id,c.customer_name,cs.total_sales from cs join customers c on cs
.customer_id=c.customer_id cross join avg_s where cs.total_sales>avg_val;
```

customer_id	customer_name	total_sales
BH-11710	Brosina Hoffman	12510.7020
IM-15070	Irene Maddox	9860.9480
PK-19075	Pete Kriz	17293.8680
TB-21520	Tracy Blumstein	9474.9720
MA-17560	Matt Abelman	8598.3220
SN-20710	Steve Nguyen	6255.9184
LC-16930	Linda Cazamias	8985.8920
RA-19885	Ruben Ausman	7664.6280
ES-14080	Erin Smith	9315.8480
KM-16720	Kunst Miller	9818.9440
KD-16270	Karen Daniels	16564.7160
TB-21520	Tracy Blumstein	9474.9720
JE-15745	Joel Eaton	13521.6300
JE-15745	Joel Eaton	13521.6300
SC-20770	Stewart Carmichael	8985.3140
CS-12400	Christopher Schild	7303.7170
GM-14455	Gary Mitchum	6157.2480
RB-19465	Rick Bensley	9430.9320
GZ-14470	Gary Zandusky	8710.3000
JM-15250	Janet Martin	6318.2440
CL-12565	Clay Ludtke	21761.0920
SC-20725	Steven Cartwright	10452.4200

5. What is the highest order value per customer?

```
mysql> select c.customer_id,c.customer_name,max(o.sales) highest_order_value
from customers c join orders o on c.customer_id=o.customer_id group by c.cu
stomer_id,c.customer_name;
```

customer_id	customer_name	highest_order_value
AA-10315	Alex Avila	3930.0720
AA-10375	Allen Arnold	499.9800
AA-10480	Andrew Allen	479.9700
AA-10645	Anna Andreadi	1323.9000
AB-10015	Aaron Bergman	341.9600
AB-10060	Adam Bellavance	4355.1680
AB-10105	Adrian Barton	9892.7400
AB-10150	Aimee Bixby	479.9700
AB-10165	Alan Barnes	304.2300
AB-10255	Alejandro Ballentine	479.9840
AB-10600	Ann Blume	899.1360
AC-10420	Alyssa Crouse	698.3520
AC-10450	Amy Cox	1793.9800
AC-10615	Ann Chong	1282.4100
AC-10660	Anna Chung	201.5840
AD-10180	Alan Dominguez	2999.9500
AF-10870	Art Ferguson	662.8400
AF-10885	Art Foster	349.9650
AG-10270	Alejandro Grove	1336.4400
AG-10300	Aleksandra Gannaway	230.2800
AG-10330	Alex Grayson	182.5500
AG-10390	Allen Goldenen	71.2800
AG-10495	Andrew Gjertsen	516.9600
AG-10525	Andy Gerbode	559.7100
AG-10675	Anna Gayman	2396.2656
AG-10765	Anthony Garverick	113.1000
AG-10900	Arthur Gainer	1023.9360